Walkthrough 3: AI research, many model providers, and working together

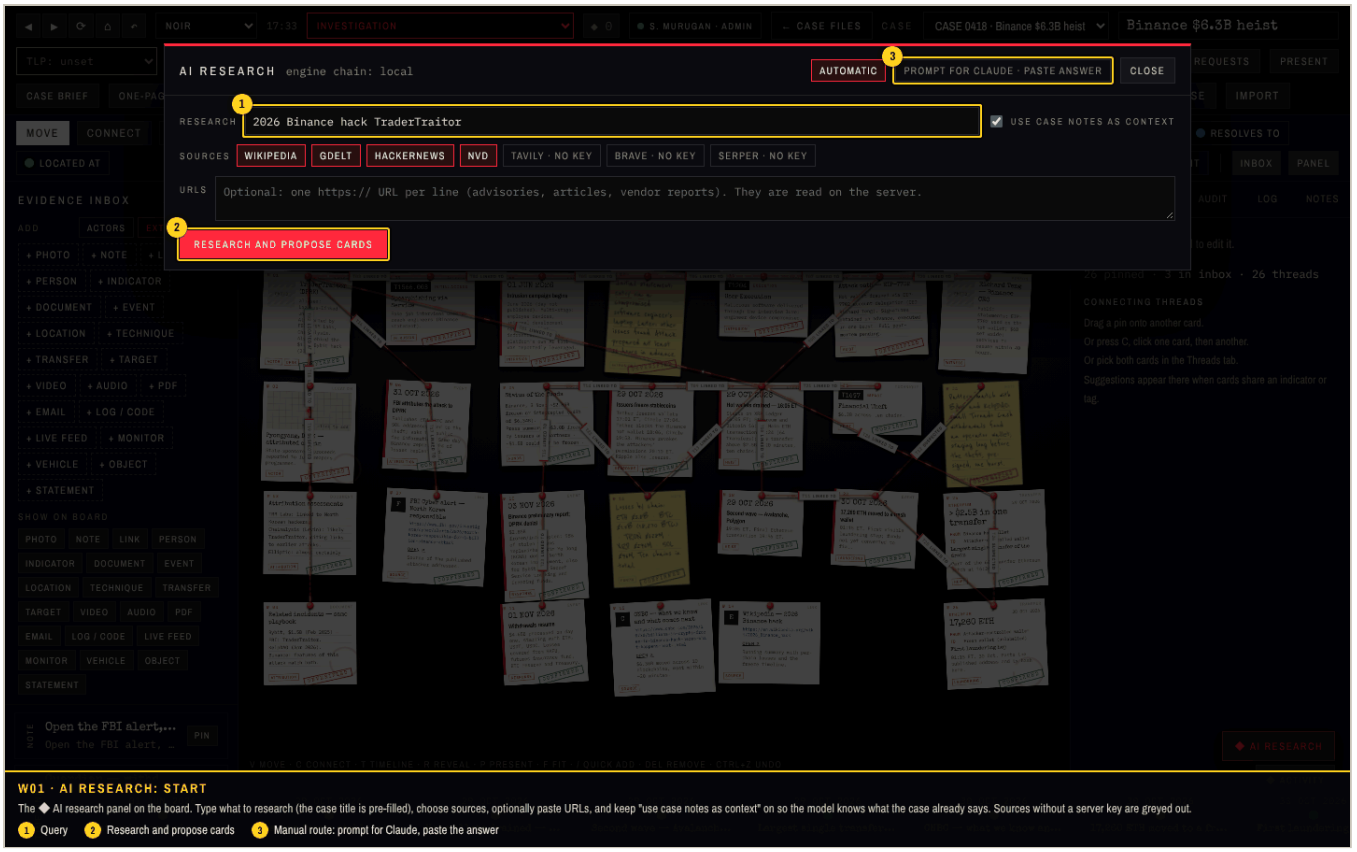
This walkthrough covers what was added after the server: the **AI research** panel that turns a query into attributed cards (with a manual route through Claude), the **multi-provider AI gateway**, and the **collaboration** layer: presence and soft locks, merged concurrent edits, revision history with restore, @mention notifications, review requests and the second-account rule for TLP:RED files. Screenshots come from `tools/capture/v3.cjs` run against a fresh server; the checks it asserts are listed at the end.

In short

Feature	Where	What it does
AI research	Board, ◆ AI research (Alt+A)	Gathers Wikipedia, GDELT news, Hacker News, NVD/KEV (and Tavily / Brave / Serper when keyed, plus pasted URLs), reads the pages on the server, proposes cards and threads with source, grade and confidence, imports what you tick into the inbox or onto the board.
Prompt for Claude, paste answer	Same panel, second tab	When the server has no model key: copies a complete prompt (instructions, case context, existing cards, sources, JSON shape) for Claude; the pasted answer is validated and imported the same way.
Model providers	Server environment	Anthropic, OpenAI, Azure OpenAI, Gemini, Cohere, Mistral, Groq, OpenRouter, DeepSeek, xAI, Together, Perplexity, Fireworks, Ollama, LM Studio, custom; tried in order with cooldown and cache; admins can list, test and disable them at runtime. The rule engine is the last resort.
Presence and soft locks	Board, top strip	Who is on the case, which exhibit they are on, who is typing in it; a notice when you select a card someone else is editing.
Merged edits	Server	A stale save is merged three ways against the revision the client started from: different fields and different cards from two analysts both survive; a deletion loses to a concurrent edit.
History	Activity drawer → History	Every save is a server revision; restore any of them, the current state stays in the history.
Notifications	Activity drawer	@mentions in comments, review requests, overdue tasks (hourly scan), watcher alerts; pushed live, with unread count; click to jump to the card.
Review requests and the TLP:RED rule	Activity drawer → Review	Send a file to a chosen reviewer; TLP:RED / AMBER+STRICT files must pass through review and be published by a different account than the requester.

Walkthrough

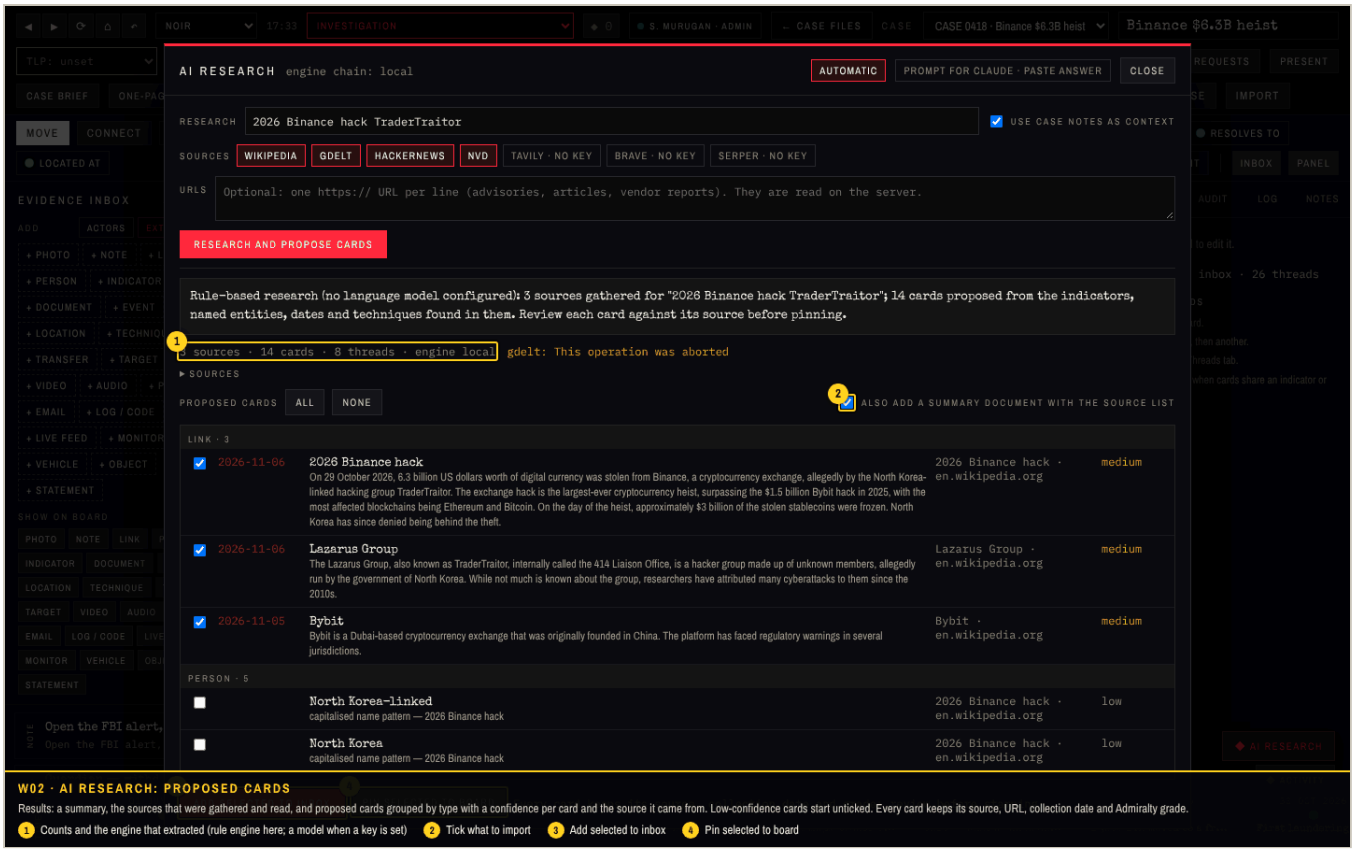
W01 · AI RESEARCH: START



The ◆ AI research panel on the board. Type what to research (the case title is pre-filled), choose sources, optionally paste URLs, and keep "use case notes as context" on so the model knows what the case already says. Sources without a server key are greyed out.

1. Query
2. Research and propose cards
3. Manual route: prompt for Claude, paste the answer

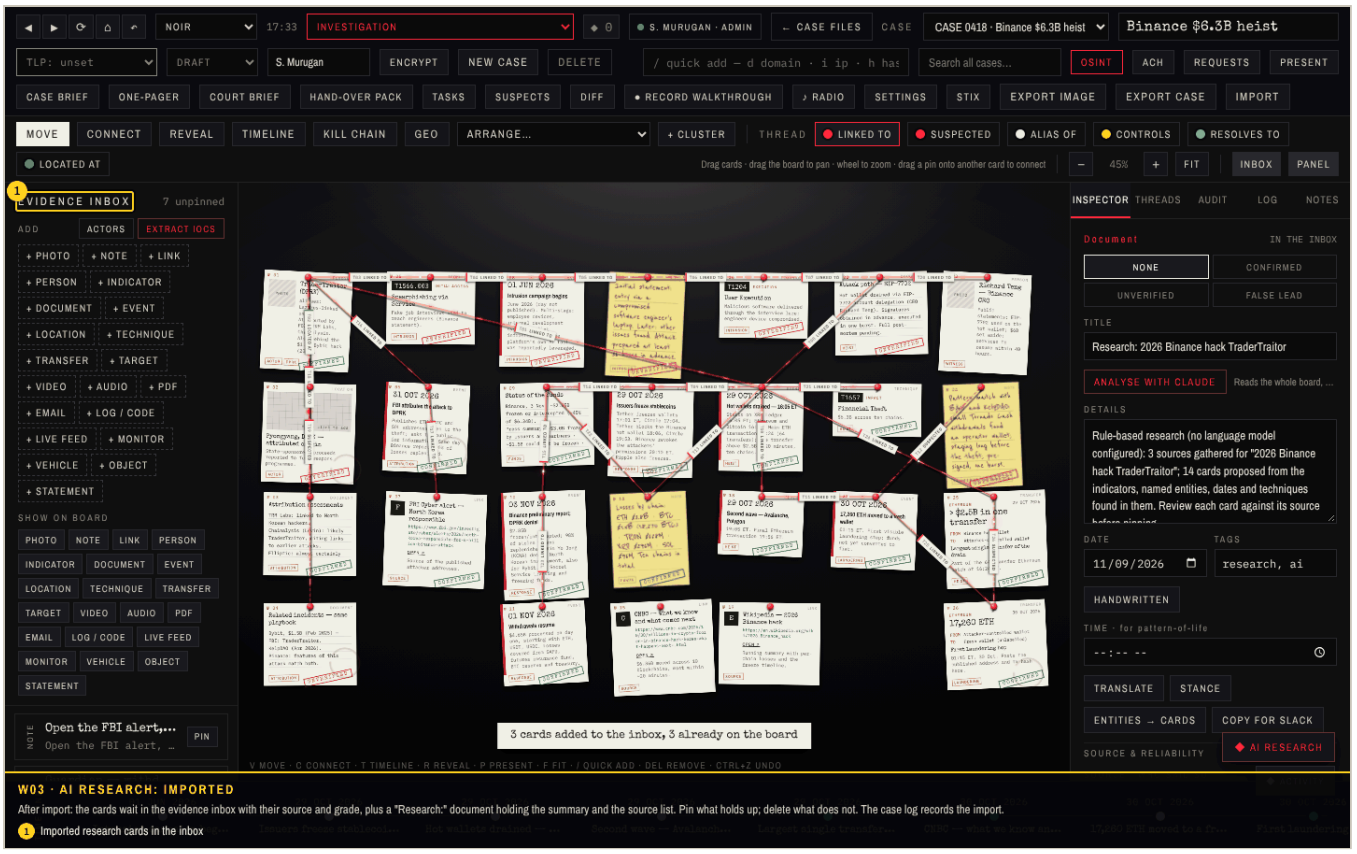
W02 · AI RESEARCH: PROPOSED CARDS



Results: a summary, the sources that were gathered and read, and proposed cards grouped by type with a confidence per card and the source it came from. Low-confidence cards start unticked. Every card keeps its source, URL, collection date and Admiralty grade.

1. Counts and the engine that extracted (rule engine here; a model when a key is set)
2. Tick what to import
3. Add selected to inbox
4. Pin selected to board

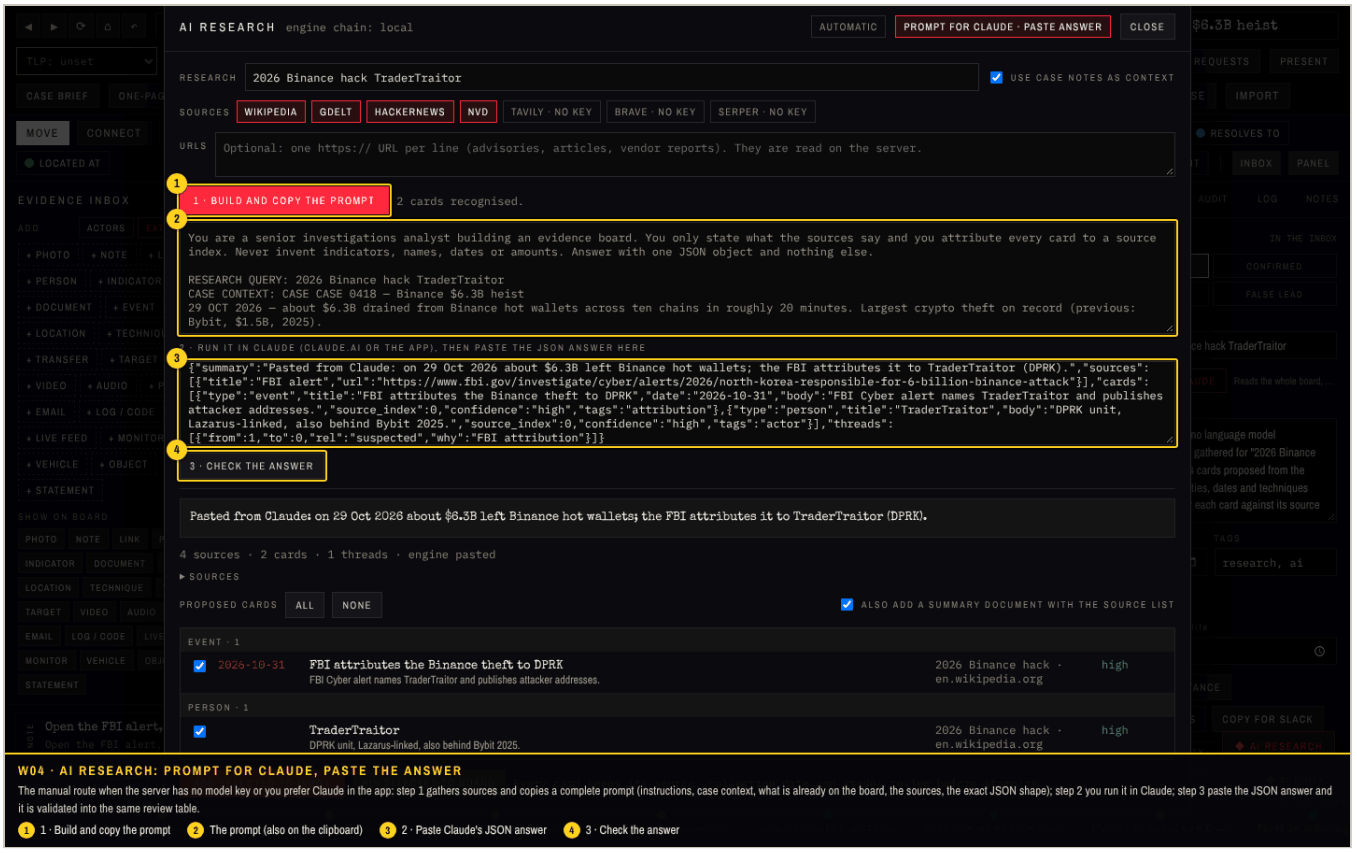
W03 · AI RESEARCH: IMPORTED



After import: the cards wait in the evidence inbox with their source and grade, plus a "Research:" document holding the summary and the source list. Pin what holds up; delete what does not. The case log records the import.

1. Imported research cards in the inbox

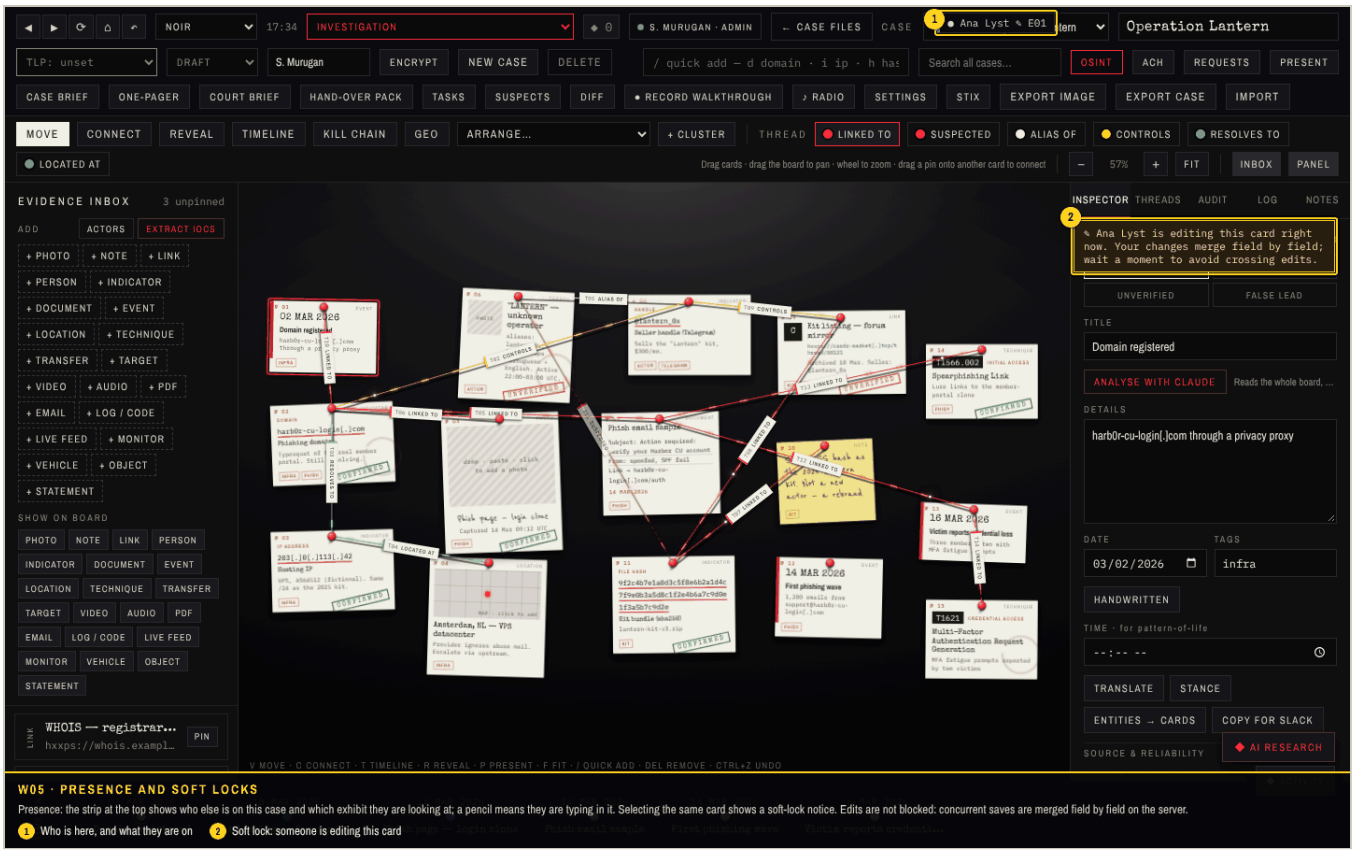
W04 · AI RESEARCH: PROMPT FOR CLAUDE, PASTE THE ANSWER



The manual route when the server has no model key or you prefer Claude in the app: step 1 gathers sources and copies a complete prompt (instructions, case context, what is already on the board, the sources, the exact JSON shape); step 2 you run it in Claude; step 3 paste the JSON answer and it is validated into the same review table.

1. 1 · Build and copy the prompt
2. The prompt (also on the clipboard)
3. 2 · Paste Claude's JSON answer
4. 3 · Check the answer

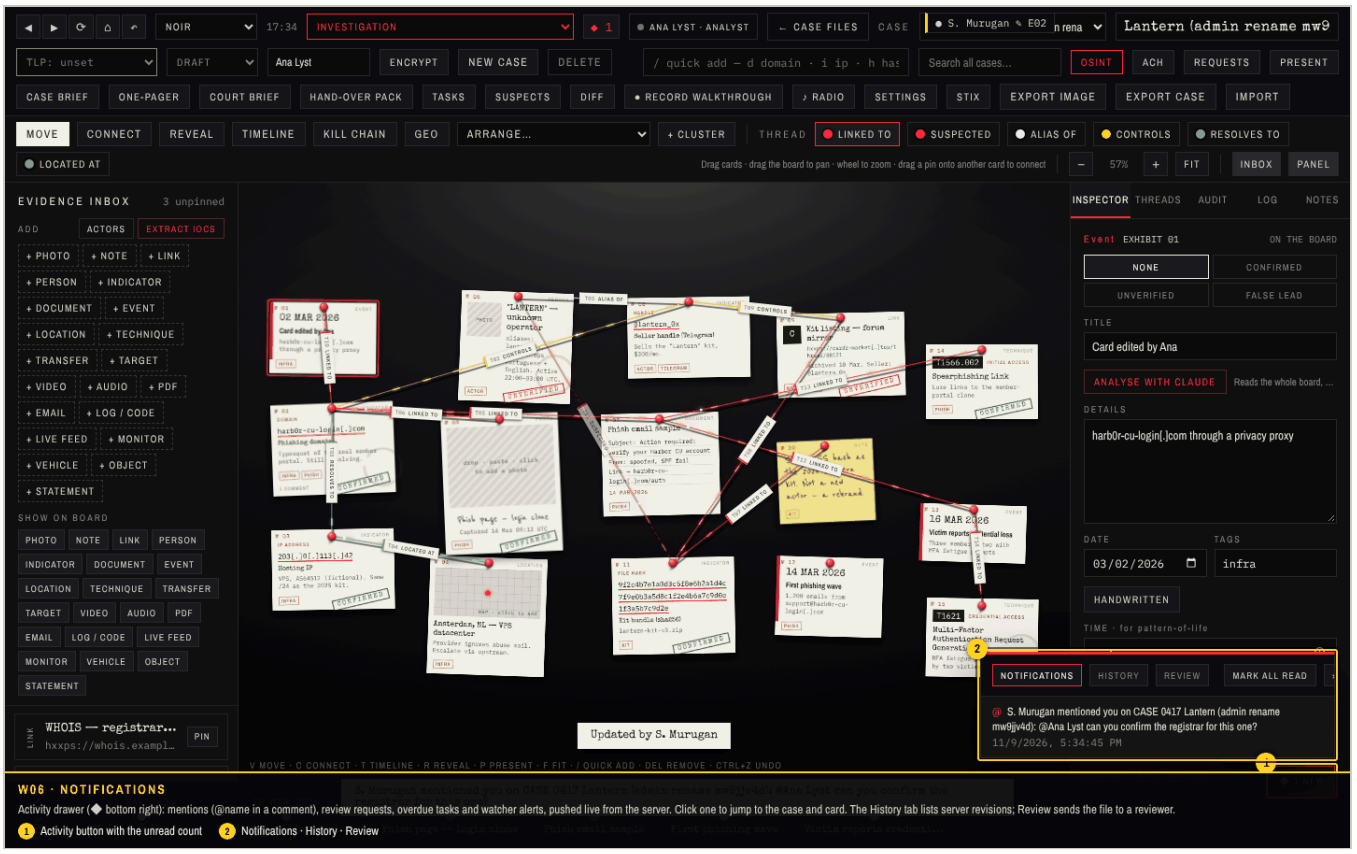
W05 · PRESENCE AND SOFT LOCKS



Presence: the strip at the top shows who else is on this case and which exhibit they are looking at; a pencil means they are typing in it. Selecting the same card shows a soft-lock notice. Edits are not blocked: concurrent saves are merged field by field on the server.

1. Who is here, and what they are on
2. Soft lock: someone is editing this card

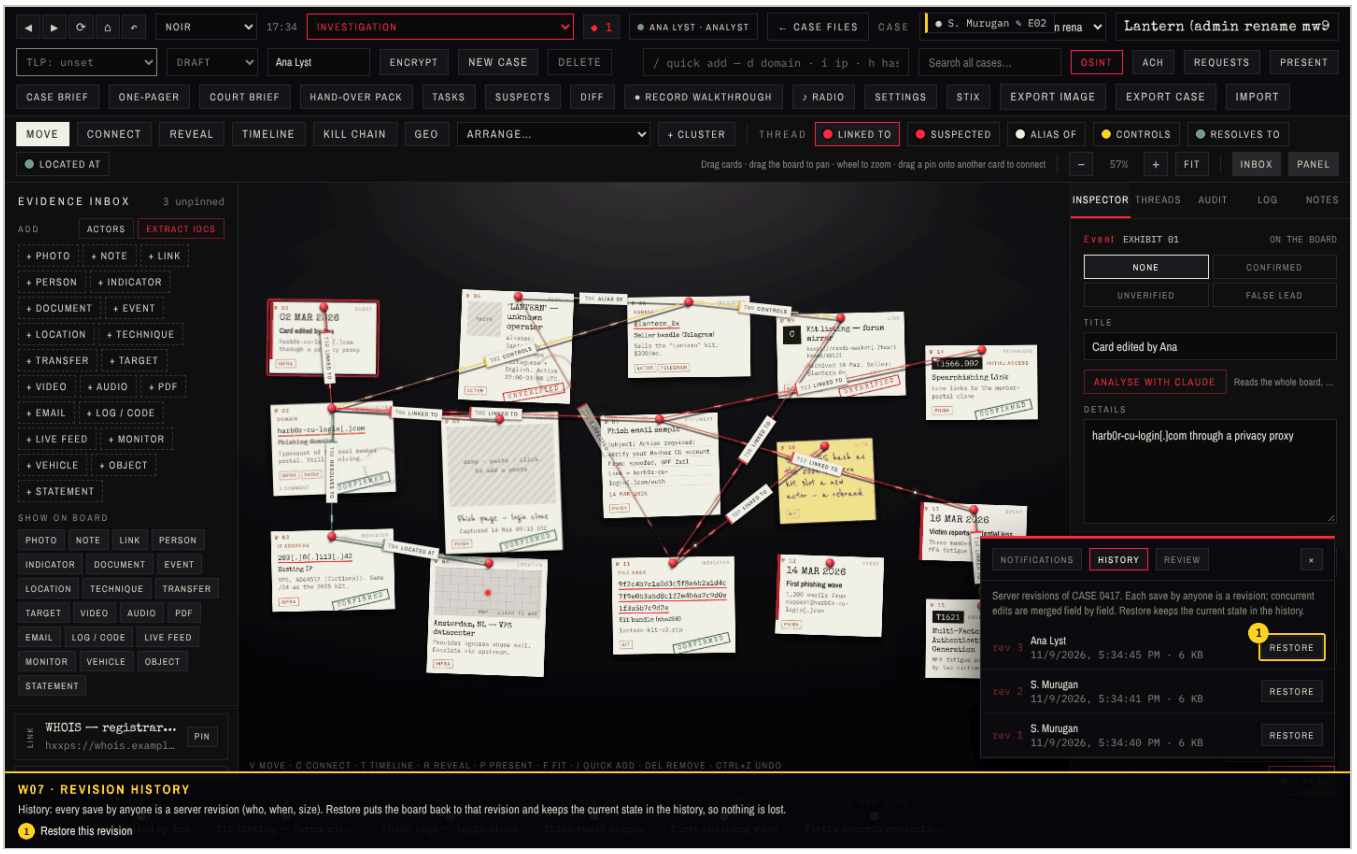
W06 · NOTIFICATIONS



Activity drawer (◆ bottom right): mentions (@name in a comment), review requests, overdue tasks and watcher alerts, pushed live from the server. Click one to jump to the case and card. The History tab lists server revisions; Review sends the file to a reviewer.

1. Activity button with the unread count
2. Notifications · History · Review

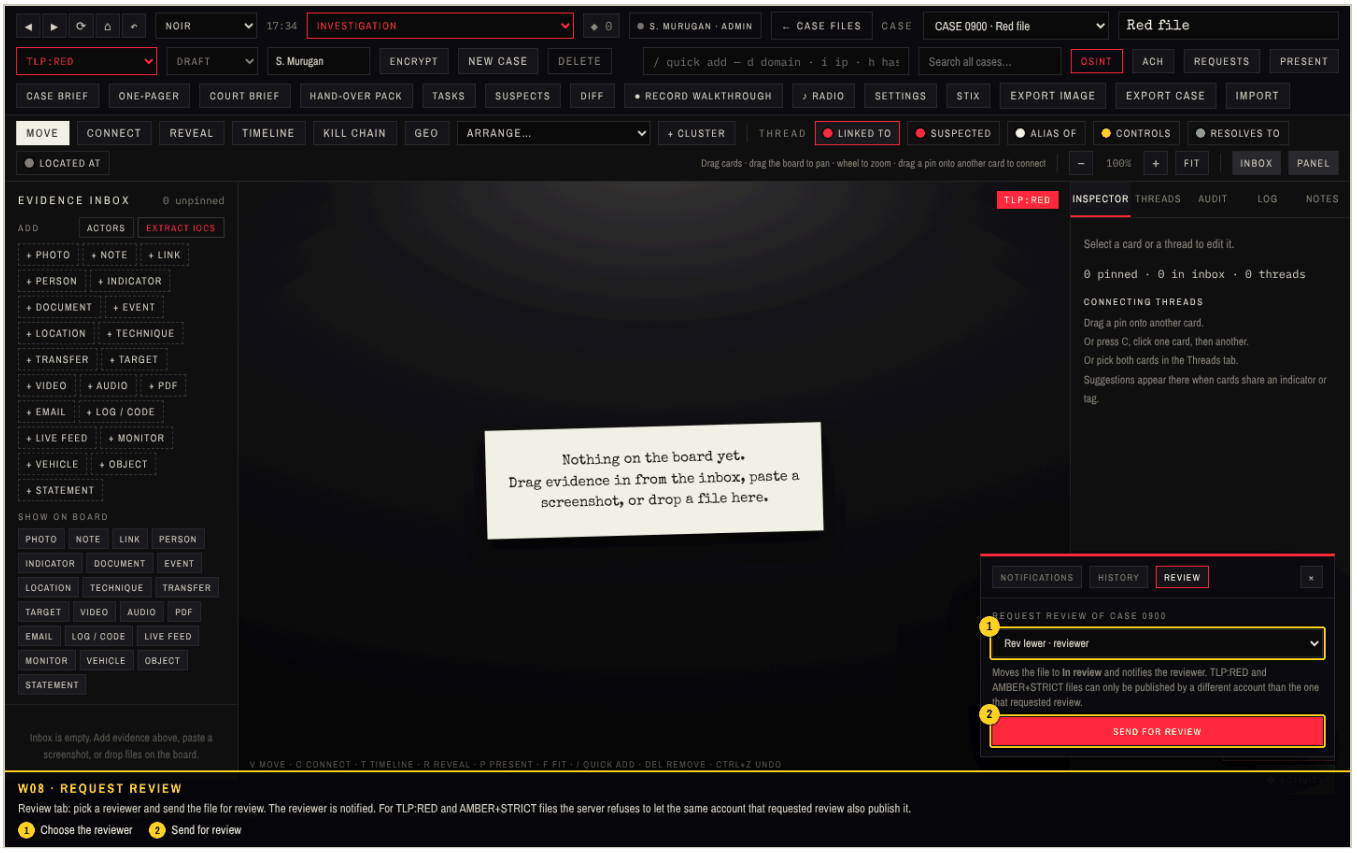
W07 · REVISION HISTORY



History: every save by anyone is a server revision (who, when, size). Restore puts the board back to that revision and keeps the current state in the history, so nothing is lost.

1. Restore this revision

W08 · REQUEST REVIEW



Review tab: pick a reviewer and send the file for review. The reviewer is notified. For TLP:RED and AMBER+STRICT files the server refuses to let the same account that requested review also publish it.

1. Choose the reviewer
2. Send for review

Checks

11 of 11 passed in the run that produced these screenshots.

Result	Check	Detail
PASS	W01 setup admin, create analyst and reviewer	
PASS	W02 the gateway reports its provider chain	local
PASS	W03 research panel: automatic research on the Binance case	3 sources · 14 cards · 8 threads · engine local
PASS	W04 import the proposed cards into the inbox	items 29 → 33
PASS	W05 manual route: build the prompt, paste a Claude answer, import	prompt 4397 chars
PASS	W06 presence: two analysts on the same case see each other; editing a card shows a soft lock	● Ana Lyst ✎ E01
PASS	W07 concurrent edits on the same case are merged, not overwritten	rev 3, 2 revisions kept
PASS	W08 @mention in a comment notifies the named analyst	1 unread for Ana
PASS	W09 history tab lists revisions and restores one	restored rev 1 → now rev 5 (Operation Lantern)

Result	Check	Detail
PASS	W10 TLP:RED: the requester cannot publish; a second account can	rejected: TLP:RED files need a second account to publish (you sent it for review)
PASS	no unexpected page errors	